

UNIT V

5781001

19. Solve $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = xe^{3x} + \sin 2x$ (11)

Or

20. Solve $x^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} + y = -\log x$. (11)

B.Tech DEGREE EXAMINATION, SEPTEMBER 2020.

First Semester

Common to all branches

MATHEMATICS-I

(From 2013-14 onwards)

Time : Three hours

Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What is the curvature of a circle?
2. Define evolute of a curve.
3. What is the chain rule of partial differentiation?
4. Write the Taylor's series for functions of two variables.
5. Find the Jacobian of $x = \frac{1}{3}(2u + v)$, $y = \frac{1}{3}(u - v)$.

6. Prove that $\int_{11}^{1410} \int_{11}^{1410} (x^2 + 4y) dy dx = 1719$.

7. Form the differential equation of simple harmonic motion given by $x = A \cos(nt + \alpha)$.

8. Write the Clairaut's equation.

9. Solve $\frac{d^4 x}{dt^4} + 4x = 0$.

10. Find the P.I of $(D^2 + 5D + 6)y = e^x$

SECTION B — (5 × 11 = 55 marks)

Answer ALL the units, choosing ONE full question from each unit.

UNIT I

11. Show that the radius of curvature of any point of cycloid $x = a(\theta + \sin \theta)$ $y = a(1 - \cos \theta)$ is $4a \cos \frac{\theta}{2}$. (11)

Or

12. Prove the following (3+3+5)

(a) $\Gamma(n+1) = n!$

(b) $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$ and find the value of $\beta\left(\frac{9}{2}, \frac{7}{2}\right)$.

UNIT II

13. Find the first and second partial derivatives of $z = x^3 + y^3 - 3axy$ and prove that $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$. (11)

Or

14. Discuss the maxima and minima of $f(x, y) = x^3 y^2 (1 - x - y)$. (11)

UNIT III

15. Change the order of integration in $I = \int_0^1 \int_{x^2}^{2-x} xy dy dx$. (11)

Or

16. Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ by changing to polar coordinates, Hence show that $\int_0^\infty e^{-x^2} dx = \sqrt{\frac{\pi}{2}}$. (11)

UNIT IV

17. Solve $x \frac{dy}{dx} + y = x^3 y^6$ (11)

Or

18. Solve $\frac{dy}{dx} - \frac{dx}{dy} = \frac{x}{y} - \frac{y}{x}$. (11)